开放经济的经济交易：核算与分析

国民收入账户和国际收支账户

- 国民收入账户记录了一国国民收入、产出和支出等经济活动的相关情况。
- 国际收支账户记录了一国对外经济交易的相关情况。

一、国民收入账户

- 国内生产总值(GDP)是一定时期，一国领土上最终商品和服务的当期生产总值。
- GDP数据表示的是按市场价格计值的最终商品和服务的当期产量。
- GDP按照支出主体分为四个部分：得出开放经济国民收入恒等式：

$$Q = C + I + G + X - M$$

- 贸易余额（净出口、贸易顺差）$NX = X - M$
- 国民生产总值GNP是一国的生产要素在一定时期内所生产的最终商品和服务的价值总量Y。
- GNP与GDP的区别：令NFP为从国外获得的净要素收入，则 $GNP = GDP + NFP$
- 如果进出口包括了要素收入:

$$EX = X + \text{从国外获得的要素收入}$$
$$IM = M + \text{向国外支付的要素收入}$$

- 则有（1-1）式:

$$Y = C + I + G + EX - IM \qquad (1-1)$$

- 其中 $CA = EX - IM$

  称为经常项目（经常账户余额）

- 经常项目与贸易余额就差了从国外获得的净要素收入(NFP): $CA = NX + NFP$

- 储蓄、投资与经常项目

  令：$S = Y - C - G$

  则：$S = I + CA$

- 注：CA的含义：度量了国际借贷的规模和方向
  - $CA < 0$:贸易赤字，借入资金，净外债增加或净国外资产减少
  - $CA > 0$:贸易盈余，贷出资金，净国外资产增加

$$CA_t = IIP_{t+1} - IIP_t \qquad IIP_t = IIP_0 + \sum_{i=0}^{t-1} CA_i$$

IIP:净外国资产(net foreign assets),也称为净国际投资头寸(net international investment position)

外部财富

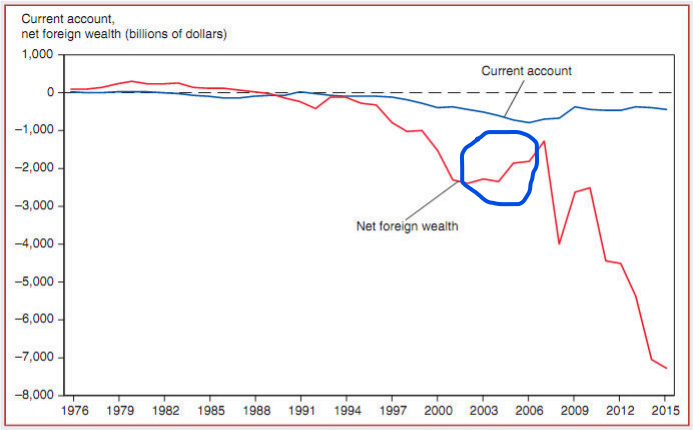
- 一个国家的净资产被称为其外部财富(external wealth)，它等于其外国资产（它拥有世界其他地区(ROW)的债权）和外国负债（它欠ROW的债务）之间的差额。

  W=ROW assets owned by home

  -Home assets owned by ROW

- 债权国(creditor nation)；债务国(debtor nation)。

美国经常项目和净外国财富余额



Pearson MyLab Economics Real-time data

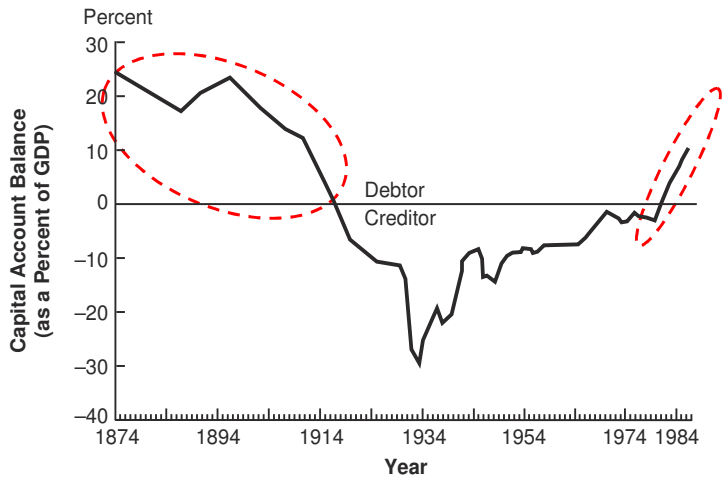
FIGURE 13-2

The U.S. Current Account and Net International Investment Position, 1976–2015

A string of current account deficits starting in the early 1980s reduced America's net foreign wealth until, by the early 21st century, the country had accumulated a substantial net foreign debt.

Source: U.S. Department of Commerce, Bureau of Economic Analysis.

美国作为国际债务人的历史



- 私人储蓄和政府储蓄:

$$S^p = Y - T - C$$

$$S^g = T - G$$

$$\therefore S^P + S^g = Y - C - G = S = I + CA$$

$$\therefore S^P = I + CA + (G - T) \qquad (1-2)$$

- 改写（1-2） $CA = S^p - I - (G - T)$
该等式为分析1980年代美日，当前美中的经常项目提供了框架。

例: 财政赤字挤出了美国制造业工人的就业?

二、国际收支账户

- 国际收支账户又称国际收支平衡表，是在特定时期对发生在一国居民与他国居民之间的货物、服务和金融资产交易的市场价值的完整记录和汇总。
- 两种类型的国际交易及其相应账户
  - 涉及商品和服务的进出口，记入经常账户
  - 涉及资产的国际交易，记入资本账户(注意！←金融账户)
- IMF《国际收支手册》第五版新增金融账户来记录原来的资产交易，而原来的资本账户用于记录特殊类别的资产流动，即非市场、非生产性资产、或无形资产，如债务减免、专利、版权、商标等。（Capital Account: flows of special categories of assets (capital): typically nonmarket, non-produced, or intangible assets like debt forgiveness, copyrights and trademarks.）

- 经常账户
  - 货物贸易
  - 服务贸易
  - 收益
  - 单方面转移
- 资本账户（←我们仍遵循习惯将金融账户称为）
  - 直接投资
  - 证券投资
  - 其他投资
- 资本账户余额（也称资本项目）KA是对外国人销售的资产与向外国人购买的资产的差额。
  - 向外国人借钱（出口资产），也称资本流入
  - 借钱给外国人（进口资产），也称资本流出

- 复式计账法则：
  - 每一笔国际交易都要以相等数额记录两次，一次记在贷方(Credits)，一次记在借方(Debits)。
  - 凡导致对外国人进行支付的交易都记入国际收支账户的借方，并在前面加（-）；凡导致从外国人那里获得支付的交易都记入国际收支账户的贷方，并在前面加（+）

- 例1：你从日本丰田公司购入一辆轿车，用$15000的支票进行支付：

	贷方	借方
经常项目（轿车进口）		-$15000
资本项目（资产出口）	+$15000	

- 例2：我国A公司向美国出口$500商品：

	贷方	借方
经常项目（商品出口）	+$500	
资本项目（资产进口）		-$500

- 例3：你在法国餐馆吃了一顿晚餐，支付$100：

	贷方	借方
经常项目（服务进口）		-$100
资本项目（资产出口）	+$100	

- 例4：你购买一股美国IBM公司发行的股票，开一张$95的支票支付股款：

	贷方	借方
资本项目（购买股票）		-$95
资本项目（支票支付）	+$95	

- 注意：这笔交易的两个方面都属于金融领域，两次记录都反映在资本账户上。如果分长短期资本账户，则借方记入的是长期资产，贷方记入的是短期资产。
- 例5：中国政府向北朝鲜提供价值$100万小麦的援助：

	贷方	借方
经常项目（出口）	+$100万	
经常项目（单方面转让）		-$100万

- 例6：美国财政部向持有美国国库券的外国居民支付$500万的利息，站在美国的角度，记为：

	贷方	借方
经常项目（收益）		-$500万
资本项目（资产出口）	+$500万	

- 例7：美国银行通过债务重组免除了巴格纳政府欠下的$5000债务。这种情况下美国银行对其进行了转移支付，记入资本账户；与此相关美国银行减少持有的海外资产，贷记金融账户。

	Credits	Debits
债务豁免 *(资本账户，U.S. 转移支付)*		$5,000
银行减少对巴格纳的债权 *（金融账户，U.S. 资产出售）*	$5,000	

总贷方=总借方

- 我们看到，在国际收支账户中记录为贷方的每一笔交易中，都有另一笔价值相等的交易记录为借方。
- 因此，总贷方=总借方→总贷方-总借方=0
- 经常账户贷方+资本账户贷方+金融账户贷方=经常账户借方+资本账户借方+金融账户借方。

→(经常账户贷方-经常账户借方)+(资本账户贷方-资本账户借方)+(金融账户贷方-金融账户借方)=0： $CA + KA + FA = 0$

- 以下忽略资本账户，并遵循习惯将金融账户称为资本账户：

- 国际收支恒等式：

$$\text{经常账户余额} + \text{资本账户余额} = 0$$

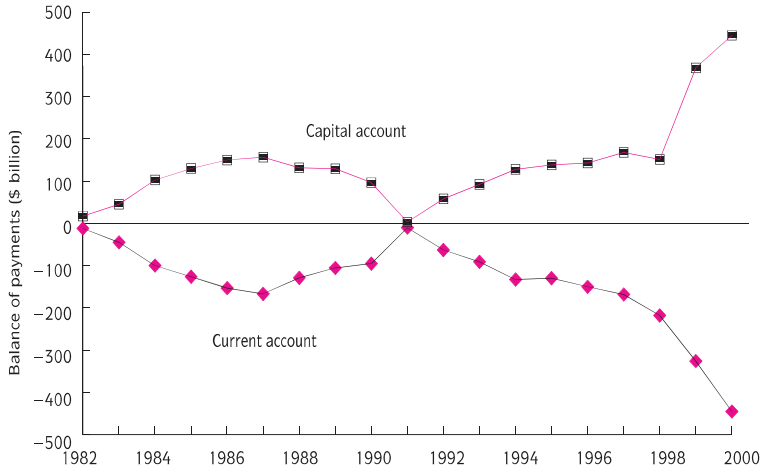
$$CA + KA = 0$$

$$\Rightarrow KA = -CA$$

- 如何理解这个恒等式？

- 问题：当一国发生经常项目逆差的时候，该国就会存在净的外国资本流入（资本项目顺差）。那怎么还会有双顺差？

美国国际收支变动



国际储备

- 国际储备是官方（央行和财政部）持有的外国资产。

- 国际储备的变动不是自主发生的，而是反映了央行的政策变化，所以将其从资本账户中划出，这样：

$$0 = CA + KA^p - \Delta R$$

或：$\Delta R = CA + KA^p$

- 称经常账户余额与私人部门资本账户余额的和(上式右边)为官方结算余额，也称国际收支余额.

- 官方储备交易：
  - 资本账户的交易中，由中央银行进行的交易称之。
  - 目的：为民间交易账户赤字融资；改变宏观经济环境
- “线上部分”和“线下部分”的含义是什么？
- 负的国际收支余额(赤字)可能是危机出现的信号。
- 为什么现实中 $CA + KA^p - \Delta R \neq 0$ ？
- 统计误差

2006年美国国际收支数据

		贷方	借方
经常账户			
1	出口	$2,096.3	
2	进口		($2,818.0)
3	单方面转移支付	$24.4	($114.0)
	经常账户余额		($811.3)
资本账户			
4	直接投资	$180.6	($235.4)
5	证券投资	$1,017.4	($426.1)
6	其他投资	$690.4	($400)
	资本账户余额	$826.9	
7	统计误差		($18)
	总余额		($2.4)
官方储备账户			$2.4

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	总余额		($2.4)
官方储备账户			$2.4

(-$811.3) + $826.9 + (-$18) = (-$2.4)

外汇储备排名

Rank	Country	Billion USD (end of *month*)
1	People's Republic of China	$ 3,305 (Mar 2012)
2	Japan	$ 1,277 (May 2012)
-	Eurozone	$ 898 (Apr 2012)
3	Saudi Arabia	$ 541 (Dec 2011)
4	Russia	$ 510 (May 2012)
5	Switzerland	$ 453 (Jun 2012)
6	Republic of China (Taiwan)	$ 395 (Apr 2012)
7	Brazil	$ 374 (Jun 2012)
8	South Korea	$ 317 (Apr 2012)
-	Hong Kong	$ 294 (Mar 2012)
9	India	$ 286 (May 2012)
10	Germany	$ 239 (May 2012)

2009年4月底，我国外汇储备排名世界经济体第一，近十年来，我国的外汇储备一直都是全球排名第一。这表明中国家底子厚了，金融实力强了吗？

家底子：对外净资产　金融实力：对外资产

世界主要国家资产负债状况

从对外资产角度来看，美国的金融实力最强

表 C4-1　2017 年末世界主要国家资产负债状况

单位：亿美元

国家（地区）	净资产	资产	负债
日本	29091	89674	60584
德国	23108	100064	76956
中国	18141	69256	51115
中国香港	13941	54690	40749
挪威	8851	17521	8669
瑞士	8691	48886	40195
新加坡	8043	36202	28160
荷兰	6123	103570	97447
沙特阿拉伯	5579	9295	3716
爱尔兰	-5409	60298	65707
法国	-5535	79296	84831
墨西哥	-5538	6031	11569
巴西	-6885	8617	15501
澳大利亚	-7711	18267	25979
西班牙	-11291	22462	33753
美国	-77250	277991	355241

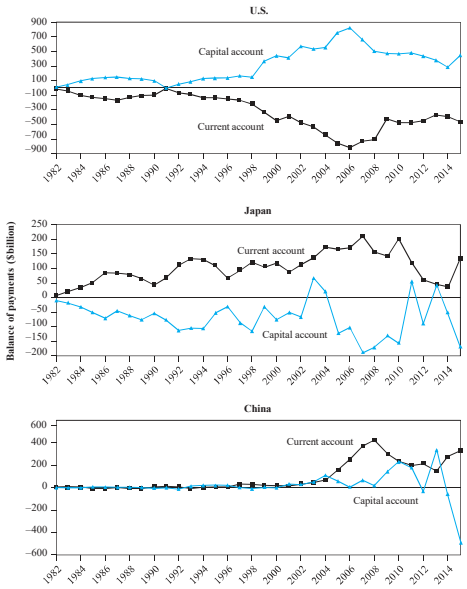
数据来源：国际货币基金组织、国家外汇管理局。

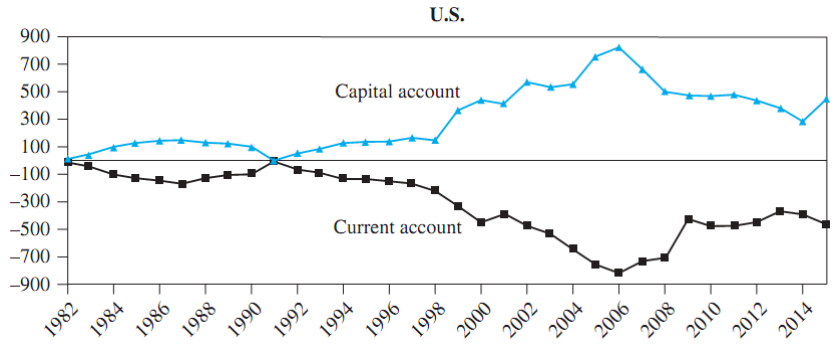
日中资本市场

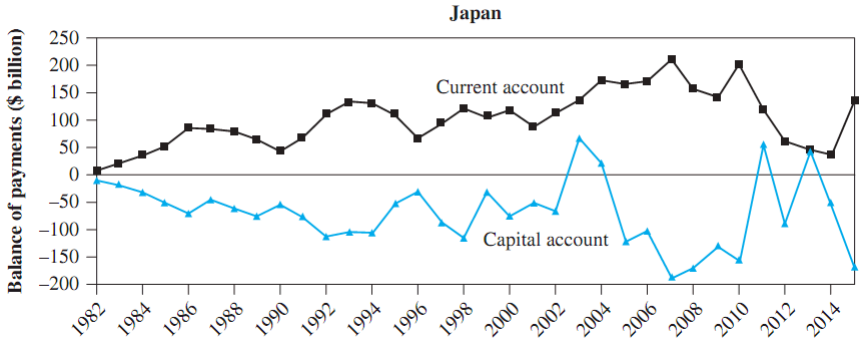
从1991年至2017年，日本一直为全球最大的净债权国。但如果只看对外投资的存量，全球海外资产最高的国家则是美国。*结合上页储备数据，国际收支数据反映出中日两国经济怎样的差别？*

外汇储备反映的是"官方的家底子"。中国"官方的家底子"厚，但私人部门不然。

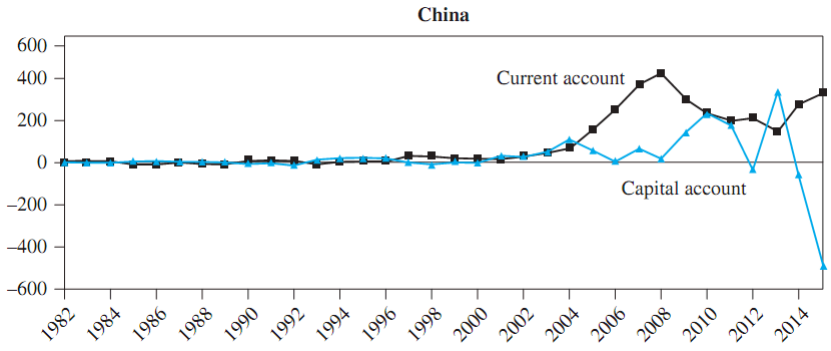
Balance-of-Payments Trends: 1982–2015



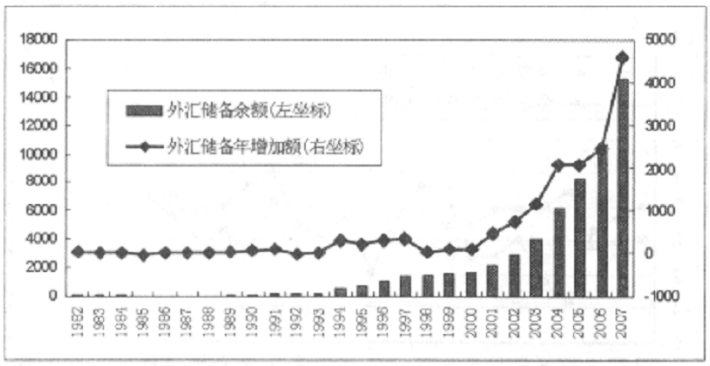

U.S.
900
700
500
300
100
−100
−300
−500
−700
−900
Capital account
Current account
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014


Japan
Balance of payments ($ billion)
250
200
150
100
50
0
−50
−100
−150
−200
Current account
Capital account
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014

双顺差：指的是CA和KA^p（私人部门的资本账户）双顺差。CA和KA不会双顺差。根据CA + KA^p + ΔR = 0可知，中国的ΔR很大。



中国官方外汇储备（1982-2007）单位:亿美元



数据来源：国家外汇管理局历年《国际收支平衡表》。

TABLE 13-2 U.S. Balance of Payments Accounts for 2012 (billions of dollars)

Current Account	
(1) Exports	**2,986.9**
Of which:	
Goods	1,561.2
Services	649.3
Income receipts (primary income)	776.3
(2) Imports	**3,297.7**
Of which:	
Goods	2,302.7
Services	442.5
Income payments (primary income)	552.4
(3) Net unilateral transfers (secondary income)	−129.7
Balance on current account [(1) − (2) + (3)]	**− 440.4**
Capital Account	
(4)	**7.0**
Financial Account	
(5) Net U.S. acquisition of financial assets, excluding financial derivatives	**97.5**
Of which:	
Official reserve assets	4.5
Other assets	93.0
(6) Net U.S. incurrence of liabilities, excluding financial derivatives	**543.9**
Of which:	
Official reserve assets	393.9
Other assets	150.0
(7) Financial derivatives, net	**7.1**
Net financial flows [(5) − (6) + (7)]	**− 439.4**
Net errors and omissions [Net financial flows less sum of current and capital accounts]	−6.0

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 14, 2013, release. Totals may differ from sums because of rounding.

Income is made up mostly of international interest and dividend payments and the earnings of domestically owned firms operating abroad.

NUT = gifts received – gifts given. Gifts given exceeded gifts received by $129.7 billion

A *negative* current account is called a *deficit*: $440.4 billion has to be made up by borrowing.

Credits – Debits

应该是(6)-(5)

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Of which:	
Goods	1,561.2
Services	649.3
Income receipts (primary income)	776.3
(2) Imports	**3,297.7**
Of which:	
Goods	2,302.7
Services	442.5
Income payments (primary income)	552.4
(3) Net unilateral transfers (secondary income)	−129.7
Balance on current account [(1) − (2) + (3)]	**− 440.4**
Capital Account	
(4)	**7.0**
Financial Account	
(5) Net U.S. acquisition of financial assets, excluding financial derivatives	**97.5**
Of which:	
Official reserve assets	4.5
Other assets	93.0
(6) Net U.S. incurrence of liabilities, excluding financial derivatives	**543.9**
Of which:	
Official reserve assets	393.9
Other assets	150.0
(7) Financial derivatives, net	**7.1**
Net financial flows [(5) − (6) + (7)]	**− 439.4**
Net errors and omissions [Net financial flows less sum of current and capital accounts]	−6.0

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 14, 2013, release. Totals may differ from sums because of rounding.

Income is made up mostly of international interest and dividend payments and the earnings of domestically owned firms operating abroad.

NUT = gifts received – gifts given. Gifts given exceeded gifts received by $129.7 billion

A *negative* current account is called a *deficit*: $440.4 billion has to be made up by borrowing.

Credits – Debits

This is credits – debits. It is a *positive* amount, which indicates that $7.0 billion was *received*.
But that still leaves $433.4 to be borrowed.

美国持有的外国资产(97.5)少于外国持有的美国资产(543.9)，但CA里的利息收入(Income receipts)大于利息支出(Income payments)，说明美国持有的外国资产回报率高。这是因为外国持有的美国资产大多数是回报率低的美国国债。

美国的国际投资头寸

- 美国拥有世界上最高的负外部财富，因此是世界上最大的债务国。
- 其经常账户持续出现赤字
- 自1980年以来，美国持有的外国资产价值一直在增长，但美国的债务（外国人持有的资产）增长得更快。

美国对外资产与负债

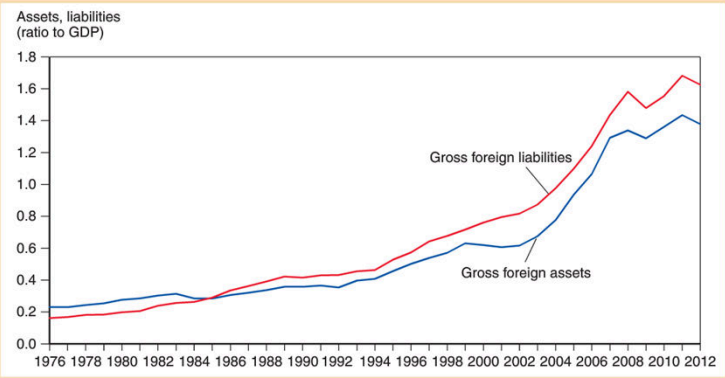

FIGURE 13-3 MyEconLab Real-time data

U.S. Gross Foreign Assets and Liabilities, 1976–2012

Since 1976, both the foreign assets and the liabilities of the United States have increased sharply. But liabilities have risen more quickly, leaving the United States with a substantial net foreign debt.

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 2013.

外部财富的变化

- 由前，一个国家的外部财富水平随时间变化的原因是CA的变化，它对应于同期的资本流动

- 但外部财富的变化还有另外一个原因：估值影响 例子见P4图片

- 以前获得外国资产的市场价格的变化可以改变一个国家的外部财富。

- 汇率变化也有类似的效果。

- 因此外部财富变化公式需要修正为：

$$\Delta IIP = \Delta NFA = CA + VAL$$

VAL：valuation effects (估值效应)

- 例如：2011-2012年，美国外部财富的变化：

  ΔNFA=(-3,863,892)-(-3,730,590)=-133,302

  CA =-439,351

  VAL=489,566+5100+(-188,618)

- 即$439,351的CA逆差被估值效应(价格和汇率的变化)抵消有余。（见下表，单位:百万$）

美国国际投资头寸表

TABLE 13-3 International Investment Position of the United States at Year End, 2011 and 2012 (millions of dollars)

Line	Type of investment	Position, 2011[r]	Changes in position in 2012: Attributable to: Financial flows (a)	Valuation adjustments: Price changes (b)	Exchange-rate changes[1] (c)	Other changes[2] (d)	Total (a+b+c+d)	Position, 2012[p]
1	**Net international investment position of the United States (lines 2+3)**	**−3,730,590**	**−439,351**	**489,566**	**5,100**	**−188,618**	**−133,302**	**−3,863,892**
2	Financial derivatives, net (line 5 less line 25)[3]	86,039	7,064	(D)	(D)	[4]−35,327	−28,263	57,776
3	Net international investment position, excluding financial derivatives (line 6 less line 26)	−3,816,629	−446,415	489,566	5,100	−153,291	−105,039	−3,921,668
4	**U.S.-owned assets abroad (lines 5+6)**	**21,636,152**	(D)	(D)	(D)	(D)	**1,466**	**21,637,618**
5	Financial derivatives (gross positive fair value)	4,716,578	(D)	(D)	(D)	(D)	−1,096,817	3,619,761
6	U.S.-owned assets abroad, excluding financial derivatives (lines 7+12+17)	16,919,574	97,469	990,880	5,909	4,024	1,098,283	18,017,857
7	U.S. official reserve assets	537,037	4,460	33,079	−2,208	0	35,331	572,368
8	Gold	400,355	0	[5]33,079	.....	[6]0	33,079	433,434
9	Special drawing rights	54,956	37	.....	57	0	94	55,050
10	Reserve position in the International Monetary Fund	30,080	4,032	.....	49	0	4,081	34,161
11	Foreign currencies	51,646	391	.....	−2,314	0	−1,923	49,723
12	U.S. government assets, other than official reserve assets	178,901	−85,331	.....	(*)	0	−85,331	93,570
13	U.S. credits and other long-term assets[7]	78,373	5,656	.....	(*)	0	5,656	84,029
14	Repayable in dollars	78,100	5,656	.....	.....	0	5,656	83,756
15	Other[8]	273	0	.....	(*)	.....	0	273
16	U.S. foreign currency holdings and U.S. short-term assets[9]	100,528	−90,987	.....	(*)	.....	−90,987	9,541
17	U.S. private assets	16,203,636	178,341	957,801	8,117	4,024	1,148,283	17,351,919
18	Direct investment at current cost	4,663,142	388,293	25,339	16,234	−15,258	414,608	5,077,750
19	Foreign securities	6,441,350	144,823	932,462	−7,412	20,000	1,089,873	7,531,223
20	Bonds	1,938,912	62,243	139,503	−973	0	200,773	2,140,685
21	Corporate stocks	4,501,438	82,580	792,959	−6,439	20,000	889,100	5,390,538
22	U.S. claims on unaffiliated foreigners reported by U.S. nonbanking concerns	792,953	25,723	.....	3,194	22,882	51,799	844,752
23	U.S. claims reported by U.S. banks and securities brokers, not included elsewhere	4,306,191	−380,498	.....	−3,899	−23,600	−407,997	3,898,194
24	**Foreign-owned assets in the United States (lines 25+26)**	**25,366,742**	(D)	(D)	(D)	(D)	**134,768**	**25,501,510**
25	Financial derivatives (gross negative fair value)	4,630,539	(D)	(D)	(D)	(D)	−1,068,554	3,561,985
26	Foreign-owned assets in the United States, excluding financial derivatives (lines 27+34)	20,736,203	543,884	501,314	809	157,315	1,203,322	21,939,525
27	Foreign official assets in the United States	5,256,358	393,922	42,110	58	0	436,090	5,692,448
28	U.S. government securities	4,235,886	314,660	−23,650	.....	0	291,010	4,526,896
29	U.S. Treasury securities	3,620,580	433,155	−21,531	.....	0	411,624	4,032,204
30	Other	615,306	−118,495	−2,119	.....	0	−120,614	494,692
31	Other U.S. government liabilities[10]	119,980	8,241	.....	58	0	8,299	128,279
32	U.S. liabilities reported by U.S. banks and securities brokers, not included elsewhere	205,973	−1,572	.....	.....	0	−1,572	204,401
33	Other foreign official assets	694,519	72,593	65,760	.....	0	138,353	832,872
34	Other foreign assets	15,479,845	149,962	459,204	751	157,315	767,232	16,247,077
35	Direct investment at current cost	2,879,531	166,411	20,385	606	−9,607	177,795	3,057,326
36	U.S. Treasury securities	1,386,274	156,385	−1,090	0	0	155,295	1,541,569
37	U.S. securities other than U.S. Treasury securities	6,151,552	196,908	439,909	−897	116,578	752,498	6,904,050
38	Corporate and other bonds	2,894,604	23,584	125,774	−897	18,898	167,359	3,061,963
39	Corporate stocks	3,256,948	173,324	314,135	.....	97,680	585,139	3,842,087
40	U.S. currency	397,086	57,141	.....	.....	0	57,141	454,227
41	U.S. liabilities to unaffiliated foreigners reported by U.S. nonbanking concerns	630,925	−39,505	.....	3,158	61,944	25,597	656,522
42	U.S. liabilities reported by U.S. banks and securities brokers, not included elsewhere	4,034,477	−387,378	.....	−2,116	−11,600	−401,094	3,633,383
	Memoranda:							
43	Direct investment abroad at market value	4,513,863	388,293	301,652	48,194	−2,463	735,676	5,249,539
44	Direct investment in the United States at market value	3,510,395	166,411	260,399	.....	−13,236	413,574	3,923,969

TABLE 13-3 International Investment Position of the United States at Year End, 2011 and 2012 (millions of dollars)

Recall from Table 13-2 that this was the US current account deficit in 2012.

Line	Type of investment	Position, 2011[r]	Changes in position in 2012: Attributable to: Financial flows (a)	Valuation adjustments: Price changes (b)	Valuation adjustments: Exchange-rate changes[1] (c)	Valuation adjustments: Other changes[2] (d)	Total (a+b+c+d)	Position, 2012[r]
1	**Net international investment position of the United States (lines 2+3)**	**–3,730,590**	**–439,351**	**489,566**	**5,100**	**–188,618**	**–133,302**	**–3,863,892**
2	Financial derivatives, net (line 5 less line 25)[3]	86,039	7,064	([4])	([4])	[4]–35,327	–28,263	57,776
3	Net international investment position, excluding financial derivatives (line 6 less line 26)	–3,816,629	–446,415	489,566	5,100	–153,291	–105,039	–3,921,668
4	**U.S.-owned assets abroad (lines 5+6)**	**21,636,152**	([3])	([3])	([3])	([3])	**1,466**	**21,637,618**
5	Financial derivatives (gross positive fair value)	4,716,578	([3])	([3])	([3])	([3])	–1,096,817	3,619,761
6	U.S.-owned assets abroad, excluding financial derivatives (lines 7+12+17)	16,919,574	97,469	990,880	5,909	4,024	1,098,283	18,017,857
7	U.S. official reserve assets	537,037	4,460	33,079	–2,208	0	35,331	572,368
8	Gold	400,355	0	[5] 33,079	.....	[6] 0	33,079	433,434
9	Special drawing rights	54,956	37	.....	57	0	94	55,050
10	Reserve position in the International Monetary Fund	30,080	4,032	.....	49	0	4,081	34,161
11	Foreign currencies	51,646	391	.....	–2,314	0	–1,923	49,723
12	U.S. government assets, other than official reserve assets	178,901	–85,331	.....	(*)	0	–85,331	93,570
13	U.S. credits and other long-term assets[7]	78,373	5,656	.....	(*)	0	5,656	84,029
14	Repayable in dollars	78,100	5,656	.....	.....	0	5,656	83,756
15	Other[8]	273	0	.....	(*)	.....	0	273
16	U.S. foreign currency holdings and U.S. short-term assets[9]	100,528	–90,987	.....	(*)	.....	–90,987	9,541
17	U.S. private assets	16,203,636	178,341	957,801	8,117	4,024	1,148,283	17,351,919
18	Direct investment at current cost	4,663,142	388,293	25,339	16,234	–15,258	414,608	5,077,750
19	Foreign securities	6,441,350	144,823	932,462	–7,412	20,000	1,089,873	7,531,223
20	Bonds	1,939,912	62,243	139,503	–973	0	200,773	2,140,685
21	Corporate stocks	4,501,438	82,580	792,959	–6,439	20,000	889,100	5,390,538
22	U.S. claims on unaffiliated foreigners reported by U.S. nonbanking concerns	792,953	25,723	.....	3,194	22,882	51,799	844,752
23	U.S. claims reported by U.S. banks and securities brokers, not included elsewhere	4,306,191	–380,498	.....	–3,899	–23,600	–407,997	3,898,194

TABLE 13-3 International Investment Position of the United States at Year End, 2011 and 2012 (millions of dollars)

These reflect changes in the dollar values of assets bought (sold) by US residents from (to) foreign residents caused by asset price changes and exchange rate changes.

Line	Type of investment	Position, 2011[r]	Financial flows (a)	Price changes (b)	Exchange-rate changes[1] (c)	Other changes[2] (d)	Total (a+b+c+d)	Position, 2012[r]
1	**Net international investment position of the United States (lines 2+3)**	**–3,730,590**	**–439,351**	**489,566**	**5,100**	**–188,618**	**–133,302**	**–3,863,892**
2	Financial derivatives, net (line 5 less line 25)[3]	86,039	7,064	([4])	([4])	[4]–35,327	–28,263	57,776
3	Net international investment position, excluding financial derivatives (line 6 less line 26)	–3,816,629	–446,415	489,566	5,100	–153,291	–105,039	–3,921,668
4	**U.S.-owned assets abroad (lines 5+6)**	**21,636,152**	([3])	([3])	([3])	([3])	**1,466**	**21,637,618**
5	Financial derivatives (gross positive fair value)	4,716,578	([3])	([3])	([3])	([3])	–1,096,817	3,619,761
6	U.S.-owned assets abroad, excluding financial derivatives (lines 7+12+17)	16,919,574	97,469	990,880	5,909	4,024	1,098,283	18,017,857
7	U.S. official reserve assets	537,037	4,460	33,079	–2,208	0	35,331	572,368
8	Gold	400,355	0	[5] 33,079	.....	[6] 0	33,079	433,434
9	Special drawing rights	54,956	37	.....	57	0	94	55,050
10	Reserve position in the International Monetary Fund	30,080	4,032	.....	49	0	4,081	34,161
11	Foreign currencies	51,646	391	.....	–2,314	0	–1,923	49,723
12	U.S. government assets, other than official reserve assets	178,901	–85,331	.....	(*)	0	–85,331	93,570
13	U.S. credits and other long-term assets[7]	78,373	5,656	.....	(*)	0	5,656	84,029
14	Repayable in dollars	78,100	5,656	.....	.....	0	5,656	83,756
15	Other[8]	273	0	.....	(*)	.....	0	273
16	U.S. foreign currency holdings and U.S. short-term assets[9]	100,528	–90,987	.....	(*)	.....	–90,987	9,541
17	U.S. private assets	16,203,636	178,341	957,801	8,117	4,024	1,148,283	17,351,919
18	Direct investment at current cost	4,663,142	388,293	25,339	16,234	–15,258	414,608	5,077,750
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22	U.S. claims on unaffiliated foreigners reported by U.S. nonbanking concerns	792,953	25,723	.....	3,194	22,882	51,799	844,752
23	U.S. claims reported by U.S. banks and securities brokers, not included elsewhere	4,306,191	–380,498	.....	–3,899	–23,600	–407,997	3,898,194

24	**Foreign-owned assets in the United States (lines 25+26)**	**25,366,742**	(3)	(3)	(3)	(3)	**134,768**	**25,501,510**
25	Financial derivatives (gross negative fair value)	4,630,539	(3)	(3)	(3)	(3)	−1,068,554	3,561,985
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27	Foreign official assets in the United States	5,256,358	393,922	42,110	58	0	436,090	5,692,448
28	U.S. government securities	4,235,886	314,660	−23,650	.....	0	291,010	4,526,896
29	U.S. Treasury securities	3,620,580	433,155	−21,531	.....	0	411,624	4,032,204
30	Other	615,306	−118,495	−2,119	.....	0	−120,614	494,692
31	Other U.S. government liabilities[10]	119,980	8,241	.....	58	0	8,299	128,279
32	U.S. liabilities reported by U.S. banks and securities brokers, not included elsewhere	205,973	−1,572	.....	.....	0	−1,572	204,401
33	Other foreign official assets	694,519	72,593	65,760	.....	0	138,353	832,872
34	Other foreign assets	15,479,845	149,962	459,204	751	157,315	767,232	16,247,077
35	Direct investment at current cost	2,879,531	166,411	20,385	606	−9,607	177,795	3,057,326
36	U.S. Treasury securities	1,386,274	156,385	−1,090	0	0	155,295	1,541,569
37	U.S. securities other than U.S. Treasury securities	6,151,552	196,908	439,909	−897	116,578	752,498	6,904,050
38	Corporate and other bonds	2,894,604	23,584	125,774	−897	18,898	167,359	3,061,963
39	Corporate stocks	3,256,948	173,324	314,135	.....	97,680	585,139	3,842,087
40	U.S. currency	397,086	57,141	.....	.....	0	57,141	454,227
41	U.S. liabilities to unaffiliated foreigners reported by U.S. nonbanking concerns	630,925	−39,505	.....	3,158	61,944	25,597	656,522
42	U.S. liabilities reported by U.S. banks and securities brokers, not included elsewhere	4,034,477	−387,378	.....	−2,116	−11,600	−401,094	3,633,383
	Memoranda:							
43	Direct investment abroad at market value	4,513,863	388,293	301,652	48,194	−2,463	735,676	5,249,539
44	Direct investment in the United States at market value	3,510,395	166,411	260,399	.....	−13,236	413,574	3,923,969

r Revised
* Less than $500,000 (+/−)
..... Not applicable

1. Represents gains or losses on foreign-currency-denominated assets and liabilities due to their revaluation at current exchange rates.
2. Includes changes due to year-to-year shifts in the composition of reporting panels, primarily for bank and nonbank estimates , and to the incorporation of more comprehensive survey results. Also includes capital gains and losses of direct investment affiliates and changes in positions that cannot be allocated to financial flows, price changes, or exchange-rate changes.
3. Financial flows and valuation adjustments for financial derivatives are available only on a net basis, which is shown on line 2; they are not separately available for gross positive fair values and gross negative fair values of financial derivatives. Consequently, columns (a) through (d) on lines 4, 5, 24, and 25 are not available.
4. Data are not separately available for the three types of valuation adjustments; therefore, the sum of all three types is shown in column (d).
5. Reflects changes in the value of the official gold stock due to fluctuations in the market price of gold.
6. Reflects changes in gold stock from U.S. Treasury sales of gold medallions and commemorative and bullion coins; also reflects replenishment through open market purchases. These demonetizations/monetizations are not included in international transactions financial flows.
7. Also includes paid-in capital subscriptions to international financial institutions and outstanding amounts of miscellaneous claims that have been settled through international agreements to be payable to the U.S. government over periods in excess of 1 year. Excludes World War I debts that are not being serviced.
8. Includes indebtedness that the borrower may contractually, or at its option, repay with its currency, with a third country's currency, or by delivery of materials or transfer of services.
9. Includes foreign-currency-denominated assets obtained through temporary reciprocal currency arrangements between the Federal Reserve System and foreign central banks. These assets are included in the investment position at the dollar value established at the time they were received, reflecting the valuation of these assets in the Federal Reserve System's balance sheet. Changes in exchange rates
10. Includes U.S. government liabilities associated with military sales contracts and U.S. government reserve-related liabilities from allocations of special drawing rights (SDRs).

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 2013.

- 美国持有的外国资产中约有70%以外币计价，而几乎所有的美国对外负债(债务)都以美元计价。(这有什么优势？其他国家可以效仿吗？) 美元贬值，外币计价的资产换成美元后的数额十分庞大，债务缩水
- 汇率的变化会影响外部财富(外国总资产减去对外总负债)的价值。

中国国际收支平衡表（单位：亿美元）

2018年	借 方	贷 方	差 额
经常项目	-28645	29136	491
1. 货物和服务	-25481	26510	1029
2. 收益	-2862	2348	-514
3. 经常转移	-302	278	-24
资本项目	-9	3	-6
	资产	负债	差 额
金融项目	-3532	4838	1306
1. 直接投资	-965	2035	1070
2. 证券投资	-535	1602	1067
3.金融衍生工具	-48	-13	-62
4. 其它投资	-1984	1214	-770
储备资产			-189
净误差与遗漏			-1602

注：1.本表根据《国际收支和国际投资头寸手册》（第六版）编制。
2.“贷方”按正值列示，“借方”按负值列示，差额等于“贷方”加上“借方”。**本表除标注“贷方”和“借方”的项目外，其他项目均指差额**。

中国国际投资头寸表（单位：亿美元）

项目	2018年末
净头寸	21301
资产	73242
1 直接投资	18990
2 证券投资	4980
2.1 股权	2700
2.2 债券	2279
3 金融衍生工具	62
4 其他投资	17530
5 储备资产	31680
5.1 货币黄金	763
5.2 特别提款权	107
5.3 国际货币基金组织的储备头寸	85
5.4 外汇储备	30727
5.5 其他储备资产	-2
负债	51941
1 直接投资	27623
2 证券投资	10964
2.1 股权	6842
2.2 债券	4122
3 金融衍生工具	60
4 其他投资	13294

投资、证券投资、其他投资（主要是
的是证券投资+其他投资。把FDI从KA^p剔除掉，就得到热钱。因为存
须要用其他相对可靠的数据反推。

案例：2004年1000亿$豪赌人民币升值?

- 将资本项目进一步细分： PFA：private financial account

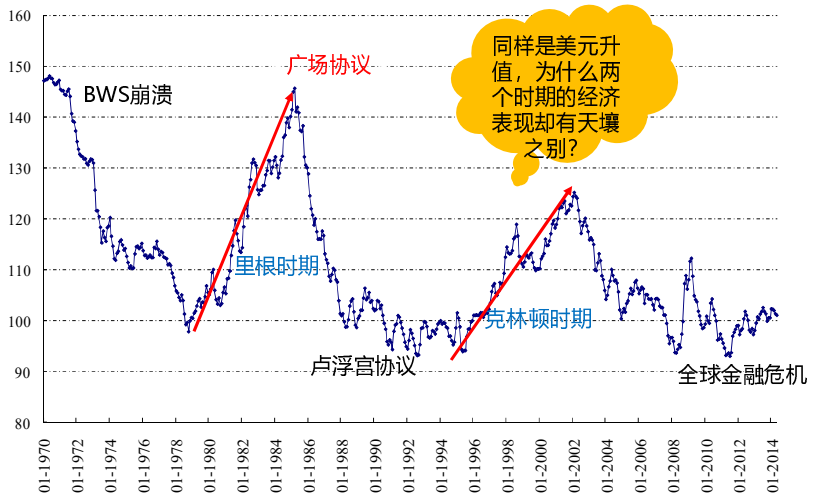
$$0 = CA + FDI - \Delta PFA - \Delta R$$

- 2004年贸易顺差：319.8亿$；
  实际利用FDI： 606.3亿$；
  外汇储备激增： 2067 亿$
- 所以 $\Delta PFA = CA + FDI - \Delta R$

$$= 319.8 + 606.3 - 2067$$

$$= -1140.9$$

USD名义有效汇率(2010=100)



贸易逆差导致制造业萧条；但资本项目为正，外国对美国的投资增加，有利于降低失业率。所以要看二者谁在主导。其中，很重要的是外国投资的流向，要流向创造就业的实体经济，而不是购买美国国债。

- 讨论1：经常项目赤字一定是坏事吗？一定导致失业增加吗？
- 分析的要点是：

$$CA = -KA$$

- 讨论2：2005年热钱流出中国了吗？

$$CA + FDI - \Delta(PFA) - \Delta(R) - miss = 0$$

仅仅是统计口径上的变化

China's Balance of Payment （billions of dollars)

	2000	2001	2002	2003	2004	2005
(1)Current account	20.52	17.41	35.42	45.87	68.66	160.82
(2)Fin & Cap acc't	1.92	34.78	32.29	52.73	110.66	62.96
(3)FDI	37.48	37.36	46.79	47.23	53.13	**67.82**
(4)=(2)-(3)	-35.56	-2.58	-14.5	5.5	57.53	**-4.86**
(5)Errors & omissions	-11.89	-4.86	7.79	18.42	27.05	-16.77
(6)=(4)+(5)	-47.45	-7.44	-6.71	23.92	84.58	-21.63
Change in foreign reserve	-10.55	-47.33	-75.51	-117.02	-206.36	-207.02

注：2005年国际收支平衡表中，外国在华直接投资流入（贷方）包括我国非金融部门和金融部门的外国在华直接投资流入。同时，根据国际货币基金组织《国际收支手册》（第五版）的原则，境内外母子和关联公司往来及贷款、以及境外机构购买建筑物亦属于直接投资统计范畴，本年也纳入统计。